

Machine Learning Project on Tiktok

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Introduction: Understanding Organic Virality

- ▶ **Context:** Short-form platforms like TikTok have redefined content consumption through the phenomenon of **virality**.
- ▶ **Research Focus:** Small to mid-tier creators (3,000 – 30,000 followers).
 - ▶ Unlike large influencers, their success relies on the **intrinsic qualities** of the content.
 - ▶ This provides an ideal setting to study **organic discovery**.
- ▶ **Key Metric: Explosion Score (S_{exp}):**
 - ▶ Measures how much a video penetrates beyond the established follower base.
 - ▶ Defined as the ratio of Views (V) to Followers (F):

$$S_{exp} = \frac{V}{F}$$

- ▶ **Objective:** Predict this score using metadata, temporal history, and multimodal signals across diverse niches.

Data Collection: Focusing on Organic Virality

- ▶ **Target:** Small creators (3,000 – 30,000 followers) to isolate content-driven success from influencer bias.
- ▶ **Strategy: Hashtag-Based Stratified Sampling:**
 1. Selected 6 niches: *Cooking, Study, Comedy, Fitness, Tech, Finance*.
 2. Collected ~3,600 videos via Apify TikTok Scraper.
 3. Filtered for active creators with sufficient posting history.
- ▶ **Final Dataset:** 466 unique creators, each with their 30 most recent videos (enabling temporal analysis).

Data Preprocessing: The Chronological Sliding Window

- ▶ **Input Construction:** Each sample integrates two distinct feature families:
 1. **Historical Performance:** Aggregated metrics from the creator's **past 10 videos** (V_{i-1} to V_{i-10}).
 2. **Current Content:** Metadata of the target video V_i available **prior to publication** to avoid data leakage.
- ▶ **Sliding Window Strategy:** Samples are constructed using a chronological sliding window, ensuring the model predicts future performance based on past behavior.
- ▶ **Evaluation Split:** A strict **chronological split** (by video rank) is used instead of random splits to prevent temporal leakage.

Feature Engineering: From Raw Metadata to Signals

1. Historical Robust Statistics

- ▶ Since views are highly skewed, we use **Median** and **Quantiles** (P70, P90) instead of the Mean to represent typical performance.
- ▶ **Normalized Engagement:** Like, comment, and share rates are calculated to decouple engagement from raw exposure.

2. Textual Feature Extraction

- ▶ **Hashtags:** Parsed into numerical indicators (*n_hashtags*, *has_fyp*, *has_viral*).
- ▶ **Captions:** Derived features including length, emoji usage, and punctuation indicators (question/exclamation marks).

Baseline & Gradient Boosting Models

Table: Performance comparison of simple baseline models & individual GBDTs

Model	R^2	MAE	Key Hyperparameters
Linear Regression (OLS)	0.1214	0.4484	N/A
Ridge	0.1346	0.4384	$\alpha = 0.0011$
KNN	0.2454	0.3667	$k = 39$, Manhattan, Distance-weighted
Random Forest	0.2750	0.3408	depth=20, estimators=100
XGBoost	0.2527	0.3461	depth=4, estimators=600, $\eta = 0.02$
LightGBM	0.2310	0.3505	leaf-wise, early_stopping=100
CatBoost	0.2526	0.3463	ordered boosting, early_stopping=100

Ensembles & The Tabular Plateau (v1 - v8)

- ▶ **Stacking Ensemble (v1):** Combined XGBoost, LightGBM, and CatBoost via a Ridge meta-learner ($R^2 = 0.3038$).

The Stacking Ensemble: A Team of Experts

The Concept (Stacked Generalization)

Combining the predictions of multiple base models with different behaviors via a supervisor "meta-learner" to obtain a more robust final prediction.

► Level 1: The Experts (Base Learners)

- **XGBoost, LightGBM, CatBoost:** Three algorithms trained on the **Training Set**.
- They generate "blind" predictions using the **Validation Set** (Out-Of-Fold): these are data points the models never saw during training.

► Level 2: The Supervisor (Meta-Learner)

- A **Ridge Regression** algorithm learns how much to trust each model.
- It performs a weighted sum of the predictions to make the final decision:

$$\hat{y} = w_{cat}P_{cat} + w_{xgb}P_{xgb} + w_{lgbm}P_{lgbm} + b$$

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- ▶ **Temporal Momentum (v2-v4):** Added sequence-aware features (short/long-term dynamics), reaching $R^2 = 0.326$.

Feature Evolution: Making the Data "Smarter" (v1-v4)

The Strategy

We moved from raw counts to "context-aware" features, helping the model understand not just the current video, but the creator's trajectory.

► **v1: The Foundations (Historical Performance)**

- `hist_p90_views`: What is the creator's "high-water mark"?
- `engagement_total_hist`: Combined historical like, comment, and share rates.

► **v2-v3: Short & Long Term Momentum**

- `momentum_3`: Are the last 3 videos doing well? (The "Hot Streak").
- `accel`: Is the growth speeding up or slowing down? ($Mom_3 - Mom_7$).

► **v4: Cross-Features (Giving the Model Context)**

- `mom3_x_tier`: Momentum multiplied by audience size.
- *Why?* Tree-based models struggle with multiplication; we explicitly defined these relationships to boost predictive power.

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- ▶ **Multimodal Integration (v5-v6):**
 - ▶ NLP (TextBlob) degraded performance due to lexicon limits.
 - ▶ Computer Vision (PIL thumbnails) improved R^2 to 0.3332.

Multimodal Integration: Giving the Model Senses

Beyond Numbers

Tabular metadata has a limit. We needed to extract signals directly from the actual content: the text and the images.

► **The Failure of Basic NLP (v5)**

- *Approach:* Used TextBlob to extract sentiment polarity and subjectivity from captions.
- *Result:* Performance dropped ($R^2 = 0.3215$).
- *Why?* Standard lexicons struggle with TikTok's short, emoji-heavy slang like "fyp" or "pov". It added noise and feature dilution, not signal.

► **The Success of Computer Vision (v6)**

- *Approach:* Processed video thumbnails into standardized 64×64 grayscale matrices, extracting brightness, contrast, and visual complexity.
- *Result:* Record performance ($R^2 = 0.3332$).
- *Why?* The visual features successfully captured genuine signal.

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- ▶ **Structural Fixes (v7):** Target encoding and top-40 feature selection confirmed a hard statistical plateau around $R^2 \approx 0.33$.

Structural Fixes: Hitting the Statistical Ceiling

Objective: Eliminate noise and instability before discarding the global model.

- ▶ **1. Top-40 Feature Selection:** Retained only the best features via a CatBoost evaluator, purging net-negative noise.
- ▶ **2. Out-Of-Fold Target Encoding:** Added the creator's historical mean. Computed strictly via a **creator-grouped 5-fold split** to prevent temporal leakage.
- ▶ **3. Exhaustive Optimization (Optuna):** Replaced random search with TPE to enforce high regularization and stable trees.

The Result: $R^2 = 0.3325$ / $MAE = 0.2922$

Maintaining the exact same score after rigorous cleaning proves the monolithic global model has hit a statistical plateau.

The Heterogeneity of Virality (v8 - v13)

- ▶ **Niche Segmentation (v8):** Breaking the global plateau via domain isolation.
- ▶ **The NLP Trap (v9-v10):** How LSA and BERT degraded the global model.
- ▶ **The Dual Pipeline (v11-v12):** Isolating spoken vs. silent content to rescue NLP.
- ▶ **Mixture of Experts (v13):** The final routing architecture and the ultimate conclusion.

Niche Segmentation: Divide and Conquer (v8)

Objective: A cooking tutorial and a comedy sketch go viral for different reasons. The global model plateaued (0.33) because it averaged out these distinct dynamics.

▶ 1. Cascade Classification (Data Sorting):

- ▶ Videos were routed into 6 niches using a strict fallback logic (Hashtags → Keywords → Creator Heritage).

▶ 2. Independent Ensembles:

- ▶ We completely isolated the training sets. A "Comedy" video was evaluated solely against other "Comedy" videos using a dedicated Stacking pipeline.

The Breakthrough

*By isolating the training data, the **Comedy segment** jumped to a record R^2 of **0.4036** and a MAE of **0.2590**.*

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The NLP Trap: When Complex AI Adds Noise (v9-v10)

Objective: We integrated advanced text analytics to capture the semantic context of subtitles, assuming it would boost global performance.

- ▶ **v9 (Speech features + LSA 30D):** Extracted "hooks" and reduced TF-IDF matrices to 30 dimensions.
- ▶ **v10 (Multilingual BERT + SVD):** Upgraded to State-of-the-Art Transformers (CamemBERT/DistilBERT).

The Statistical Backfire

- ▶ **v9 Global R^2 :** Dropped to 0.3176.
- ▶ **v10 Global R^2 :** Dropped further to 0.3080.

Conclusion: Forcing NLP features onto a dataset where 65% of videos are silent (music/visual only) confuses the Tree algorithms, diluting the structural signals.

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The Dual Pipeline: Rescuing the Semantic Signal (v11-v12)

Objective: Stop punishing the silent videos. We split the architecture entirely based on the presence of speech.

- ▶ **v11 (Dual Pipeline):**

- ▶ *Silent Pipeline:* Evaluated strictly on tabular features.
- ▶ *Spoken Pipeline:* Evaluated on Tabular + BERT Embeddings.
- ▶ **Result:** The Spoken segment recovered to an R^2 of 0.3490.

- ▶ **v12 (+ Creator Features & Niche TE):**

- ▶ We added niche-aware Target Encoding and creator profiling to the dual pipelines.
- ▶ **Result:** The Spoken segment improved further to $R^2 = 0.3693$.

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Mixture of Experts & Final Conclusion (v13)

Architecture: A master router that directs incoming videos to specialized Ensembles (Comedy Expert, Spoken Expert, Silent Expert).

- ▶ **Local Excellence vs. Global Chaos:**

- ▶ The isolated **Comedy Expert** remained highly robust ($R^2 = 0.3895$).
- ▶ However, reassembling all predictions into a **Global Assembled** metric resulted in a sharp drop ($R^2 = 0.2960$).

Progression of Test Performance

Table: Consolidated progression of test performance across all modelling iterations (v1–v13)

Step	Key change	R^2	MAE
1	Optimized XGBoost Baseline (19 features)	0.2527	0.3461
2	Stacking Ensemble [v1] : LGBM + CatBoost + Ridge L2	0.3020	0.3030
3	+ Momentum [v2] (30 features, shift-1 lag)	0.3260	0.2960
4	+ Momentum [v3] (36 features, extended windows)	0.3250	0.2970
5	+ Cross features + meta-XGB + slow lr (44 feat.) [v4]	0.3260	0.2960
6	+ NLP layer: TextBlob sentiment (51 feat.) [v5]	0.3220	0.3010
7	+ CV layer: PIL thumbnails (57 feat.) [record v6]	0.3332	0.2968
8	+ Target Enc OOF + Feature Selection + Optuna [v7]	0.3325	0.2922
9	Niche-Aware Stacking (Comedy segment) [v8, local]	0.4036	0.2590
10	+ Speech features + LSA 30D [v9, global]	0.3176	0.2951
11	LSA $\rightarrow$ BERT-SVD 20D + Hybrid Model [v10]	0.3080	0.2955
12	Dual Pipeline: Silent (tabular) + Spoken (BERT) [v11]	0.3490 [†]	0.2953
13	+ Creator features + Niche-aware TE [v12]	0.3693 [†]	0.3001
14	Mixture of Experts Routing Pipeline (Comedy Expert) [v13]	0.3895[‡]	0.3158
15	Mixture of Experts Routing Pipeline (Global Assembled) [v13]	0.2960	0.3300

[†] R^2 reported for Pipeline B (spoken segment only).

[‡] R^2 reported for specialized Comedy Expert under MoE routing.

Multi-Modal MLP: Beyond Tabular Features

Architecture: Dual-Branch MLP

► Numerical Branch:

Processes 10 core metadata/history features.

► Textual Branch: Uses *MiniLM-L12* transformer for captions, hashtags, and subtitles.

► Bottleneck Projection: Distills 384-dim text embeddings into 4-dim to prevent "signal drowning".

Key Results:

► R^2 Score: 0.3029.

► Improvement: +25.16% over the XGBoost baseline.

► Loss Function: Huber Loss ($\delta = 1.0$) for robustness against viral outliers.

Binary Classification: Viral vs. Normal

Personalized Success Labeling (Self-Baseline)

To isolate content quality from account size, success is defined relative to a creator's own history. A video is **Viral** ($y_i = 1$) if its views exceed the creator's **70th percentile** (P_{70}):

$$y_i = \begin{cases} 1 & \text{if } V_i > \text{Hist_P70_Views}_i \\ 0 & \text{otherwise} \end{cases}$$

Model Performance (24 Structural Features):

Model	Accuracy	F1-Macro	AUC-ROC
Logistic Regression	0.5064	0.4800	0.6064
KNN Classifier	0.6352	0.5268	0.5479
Random Forest	0.6599	0.5571	0.6309

► **Class Distribution:** ~29.02% Viral samples.

Three-Class Classification & The Accuracy Paradox

Multi-tier Performance Definitions:

- ▶ **Class 0 (Underperforming):** $V_i < \text{Median}$ (55.55% of samples).
- ▶ **Class 1 (Consistent):** $P_{50} \leq V_i \leq P_{70}$ (15.43% of samples).
- ▶ **Class 2 (Viral):** $V_i > P_{70}$ (29.02% of samples).

Model	Accuracy	F1-Macro	AUC-ROC
Logistic Regression	0.3959	0.3432	0.5837
KNN Classifier	0.6384	0.3808	0.5486
Random Forest	0.4861	0.4136	0.6195

The Accuracy Paradox

KNN's high accuracy is misleading—it suffers from **majority-class bias**, correctly recalling only $\sim 3.8\%$ of "Consistent" videos.

Random Forest is more robust for identifying actual performance shifts.

Conclusion: Decoding the TikTok Algorithm

- ▶ **The Statistical Plateau:** Global models reach an irreducible error limit ($R^2 \approx 0.33$) when treating all content as a single distribution.
- ▶ **Semantic Depth Matters:** Transitioning from lexical proxies (word counts) to deep multi-modal embeddings (BERT/MLP) provided a **25% performance boost** over traditional baselines.
- ▶ **Niche-Awareness is Key:** Virality is not a universal equation. The "Distributional Mixing Problem" requires specialized routing.
 - ▶ *Comedy* is momentum-driven and highly deterministic ($R^2 \approx 0.40$).
 - ▶ *Finance/Tech* are heavily influenced by exogenous trends.
- ▶ **Strategic Takeaway for Creators:** Organic growth is optimized by tailoring content to **niche-specific behavioral dynamics** (visuals, speech, and timing) rather than chasing platform-wide metadata hacks.

Classification Conclusion: Strategic Utility

- ▶ **Personalized Success Strategy:** Using a **Self-Baseline** (comparing to a creator's own P_{70} history) effectively isolates content quality from audience size bias.
- ▶ **The Accuracy Paradox (3-Class Task):**
 - ▶ **KNN Classifier** achieved the highest nominal accuracy (0.6384), but it was **biased toward the majority class** (underperforming videos).
 - ▶ **Random Forest** is practically superior (AUC-ROC ≈ 0.62): it correctly identifies performance shifts (Class 1 & 2) that creators actually care about.
- ▶ **The Metadata Ceiling:** Structural features (duration, posting time) reach a predictive plateau at **AUC** ≈ 0.62 .
- ▶ **Actionable Insight:** For true viral prediction, shifting from "exact numerical regression" to "categorical propensity" (Viral vs. Normal) provides more robust decision support for creators.

Thank You!

Any Questions?

Machine Learning Project: TikTok Organic Virality